

CAROLINE JAREB

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CAREER OBJECTIVE: Highly motivated programmer with a strong foundation in analytics, problem-solving, and communication skills developed through diverse technical, mathematical, and creative experiences. Seeking full time employment to contribute to innovative solutions that drive organizational success and contribute to a better society while fostering my professional growth. Proven ability to effectively convey complex information, foster collaboration, and drive successful outcomes through strong interpersonal connections, continuous learning and detailed planning.

EDUCATION

University of Virginia, Charlottesville, VA, *Cumulative GPA 3.6* *Graduated May 2025*

B.A. Double Major Mathematics and Computer Science (*GPA 3.9*)

Relevant Coursework: Data Structures and Algorithms, Computer Systems and Organization, Software Development Essentials (SDE), Cybersecurity, Probability, Mathematical Statistics, Discrete Mathematics, Basic Real Analysis

- Applied learned knowledge about data structures and algorithms in individual and group projects in Python, Java, and C
- Designed and implemented full-stack projects (front-end, back-end) using Test Driven Development and Version Control (SDE)

DIS Copenhagen, Semester Abroad in Copenhagen, Denmark

Fall 2023

Relevant Coursework: Artificial Intelligence, Video Game Development

SKILLS

- Proficient in Python, Java, C, SQL, Unix
- Experience with C#, JavaScript, HTML, CSS, R
- Git, Visual Studio Code
- Microsoft Office Suite and Google Workspace

RELEVANT EXPERIENCE

Biocomplexity Institute at UVA, Charlottesville, VA

June 2024 – May 2025

Undergraduate Research Assistant (August 2024 - Present)

- Visualized geospatial and weather data of large synthetic population datasets using PostgreSQL and Python in Linux Environment
- Employed UVA's High Performance Computing System to process datasets in parallel (Pandas, Dask, Python Multiprocessing)
- Researched unfamiliar frameworks such as Dask write and debug efficient code and optimize SQL queries
- Diagnosed data quality issues and implemented solutions using testing frameworks and building tools in Jupyter Notebook
- Analyzed the relationship between climate and health outcomes to ultimately write findings in a technical paper

Computing for Global Challenges Intern (June 2024 - August 2024)

- Developed reusable code in Python and JavaScript to visualize massive networks of simulated epidemic transmission networks
- Studied modeling and visualization techniques by speaking with experts and reading academic papers to contribute relevant work
- Presented results at weekly meetings to report progress as well as debug and navigate challenges with mentors and other interns
- Summarized relevant technical concepts and the results of our work in a poster and presented to other researchers

University of Virginia Department of Computer Science, Charlottesville, VA

January 2023 – May 2024

Teaching Assistant for Discrete Mathematics and Theory I (CS 2120)

- Wrote and graded weekly assignments on set theory, proof writing, logic, relations, etc.
- Discussed technical concepts and proofs with students and other TAs in weekly office hours and meetings
- Gained familiarity and knowledge of Gradescope and SIS uses and challenges as an instructor and student

ADDITIONAL EXPERIENCE

Live Arts, Charlottesville, VA

November 2024 – June 2025

Assistant Production Manager (November 2024 - Present)

- Plan a new works festival with 20+ plays and events by organizing schedules and documents in spreadsheets, attending weekly production meetings, and facilitating communication between remote and on-site playwrights, directors, and technicians
- Build rapport with members of the festival as well as community members and sponsors to gain support and host events

University of Virginia Department of Drama, Charlottesville, VA

January 2024 – May 2025

Lighting Production Assistant (January 2024 – Present)

- Followed safety protocols and communicated frequently with coworkers to cable and hang light plots with over 100 instruments
- Learned aspects of theatrical lighting hands-on: parts and functions of instruments, programming cues, and design

First Year Players (FYP) at UVA, Charlottesville, VA

August 2022 – November 2024

Producer (November 2023 - November 2024)

- Acted as primary representative of FYP (official university organization) and oversaw Production, Technical, and Artistic Staff
- Coordinated all aspects of FYP and maintained the organization's vision to foster community and create quality musical theater
- Scheduled and led meetings with staff and the entire organization, collected feedback, and implemented relevant changes